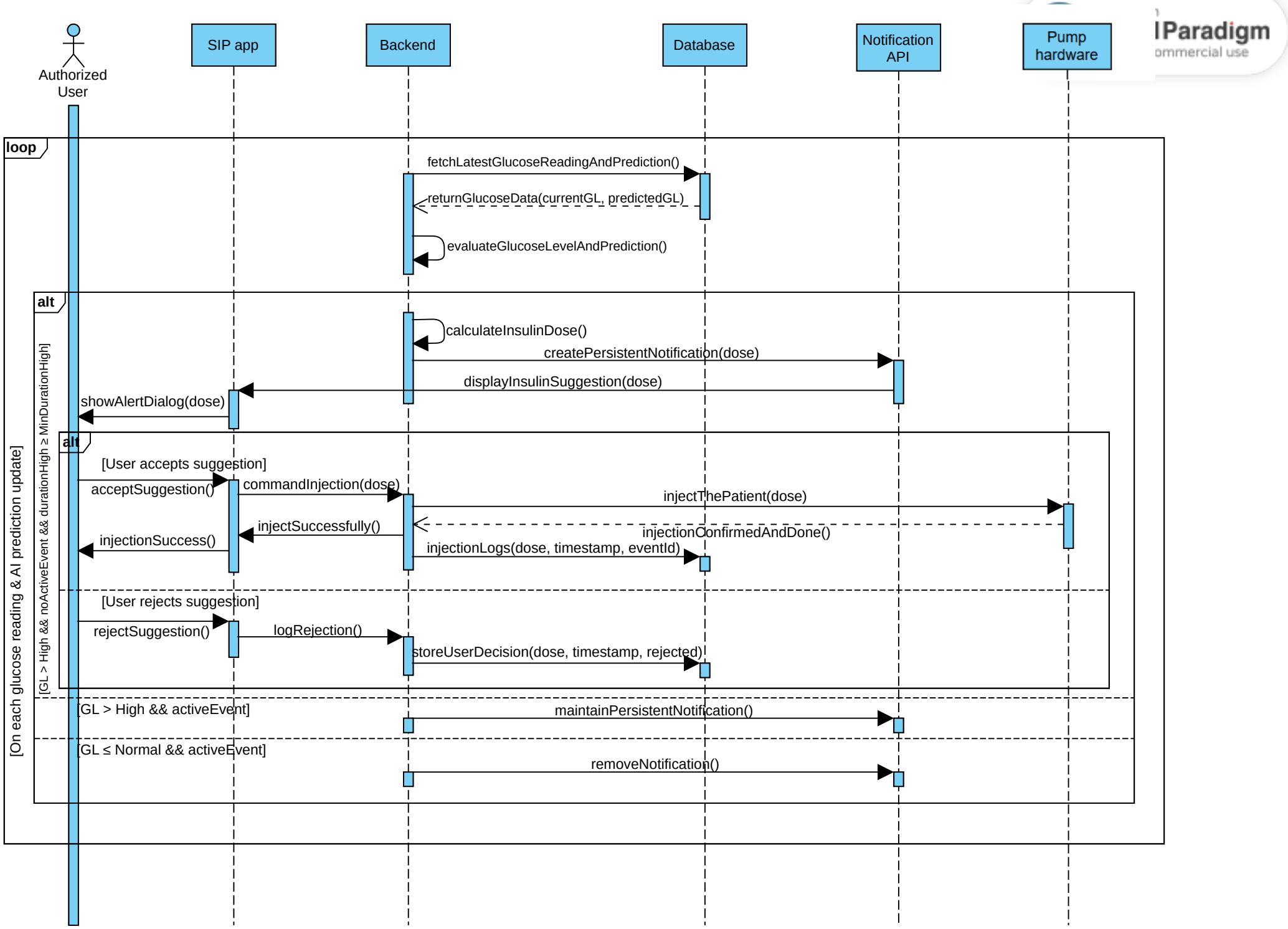
Authorized User
SIP app
Backend
Database
Notification API
Pump hardware
loop
fetchLatestGlucoseReadingAndPrediction()
returnGlucoseData(currentGL, predictedGL)
evaluateGlucoseLevelAndPrediction()
alt
calculateInsulinDose()
createPersistentNotification(dose)
displayInsulinSuggestion(dose)
showAlertDialog(dose)
alt
[User accepts suggestion]
acceptSuggestion()
commandInjection(dose)
injectThePatient(dose)
injectionConfirmedAndDone()
injectSuccessfully()
injectionSuccess()
injectionLogs(dose, timestamp, eventId)
[User rejects suggestion]
rejectSuggestion()
logRejection()
storeUserDecision(dose, timestamp, rejected)
[GL > High && noActiveEvent && durationHigh ≥ MinDurationHigh]
[GL > High && activeEvent]
maintainPersistentNotification()
[GL ≤ Normal && activeEvent]
removeNotification()
[On each glucose reading & AI prediction update]